Notes: Denis, Denis, and Yost (JF, 2002)
Global Diversification, Industrial Diversification, and Firm Value

Contents

1 Big picture

- **Research question.** Does global diversification create or destroy shareholder value, and how does it relate to industrial diversification?
- **Main idea.** U.S. firms can diversify across business lines, across national markets, or both. The paper asks whether globalization creates a value-enhancing substitute for traditional conglomerate diversification or whether it carries similar agency and coordination costs.
- **Empirical setting.** The paper studies 44,288 firm-year observations for U.S. corporations from 1984 to 1997, using Compustat industrial and geographic segment data.
- **Main trend result.** Global diversification increases over time. In the 1984 cohort, the fraction of firms that are globally diversified rises from 0.29 in 1984 to 0.45 in 1997, and foreign sales as a share of total sales among multinationals rises from 0.21 to 0.29.
- **Substitution result.** Global diversification does not substitute for industrial diversification at the individual-firm level. The two forms of diversification are positively correlated in levels and changes.
- **Main valuation result.** Global diversification is associated with a valuation discount roughly comparable to the discount for industrial diversification. In the main multivariate specification, the discount is about 0.18 for global-only diversification, 0.20 for industrial-only diversification, and 0.32 for firms that are both globally and industrially diversified.
- **Economic takeaway.** Even though global diversification became more feasible and more common, the average multinational firm does not appear to create value for shareholders relative to focused domestic benchmarks.

2 Incremental contribution of the paper

- **Moves beyond industrial diversification.** Earlier work focused heavily on conglomerates and industrial diversification. Denis, Denis, and Yost extend the diversification-discount literature to global diversification.
- **Jointly studies two diversification margins.** The paper does not treat global and industrial diversification separately. It asks whether they are substitutes, complements, or independent corporate strategies.
- **Documents time-series trends.** The paper provides evidence that global diversification rises during the same period in which industrial diversification generally falls.
- **Uses a broad panel.** The sample covers thousands of U.S. firms over 14 years and combines Compustat Industry Segment and Geographic Segment data.

- **Extends excess-value methods.** The paper adapts the Berger and Ofek excess-value framework to compare globally diversified firms with portfolios of single-segment domestic firms in the same industries.
- **Distinguishes levels and changes.** The paper studies both cross-sectional valuation discounts and within-firm changes in excess value when firms enter or exit global or industrial diversification.
- **Agency interpretation.** The findings suggest that global diversification can generate private benefits for managers, such as size, prestige, compensation, and personal risk reduction, while imposing costs on shareholders.

3 Theory and economic mechanism

3.1 Potential benefits of global diversification

The paper reviews several reasons global diversification could raise firm value.

- **Internalizing intangible assets.** Firms with information-based assets, management skills, marketing capabilities, or production know-how may gain by deploying these assets across countries.
- **Operating flexibility.** Multinational firms may shift production, sourcing, distribution, or financing across countries in response to relative prices, taxes, exchange rates, or institutional differences.
- **Investor diversification preferences.** If firms can diversify internationally more cheaply than investors, shareholders may pay a premium for multinational firms.
- **Market access.** Foreign operations may help firms reach new customers and exploit growth opportunities outside the United States.

3.2 Potential costs of global diversification

The paper also emphasizes reasons global diversification can reduce value.

- **Organizational complexity.** Multinational firms are harder to coordinate and monitor than purely domestic firms.
- **Information asymmetry inside the firm.** Headquarters may have less information about geographically distant managers and projects.
- **Cross-subsidization.** Like industrial conglomerates, multinational firms may subsidize weak units using cash flows from stronger units.

- **Managerial private benefits.** Managers may value the power, prestige, compensation, and risk-reduction benefits of running a larger multinational firm even if shareholders do not benefit.

3.3 Relation to industrial diversification

The paper sets up two broad hypotheses about the relation between global and industrial diversification.

- **Substitution hypothesis.** Global competition and product-market integration may push firms to focus on core industries while expanding internationally. If so, global diversification should be negatively related to industrial diversification.
- **Complement hypothesis.** Managers who like larger, more complex organizations may pursue both forms of diversification. If so, global and industrial diversification should be positively related.

The empirical evidence favors the complement view rather than the substitution view.

4 Data and measurement

4.1 Segment data

The data come from Compustat's Industry Segment and Geographic Segment files. Beginning in 1977, U.S. firms had to report audited financial information for industrial and foreign segments that account for more than 10 percent of consolidated sales, profits, or assets.

- **Sample period.** 1984–1997.
- **Initial sample.** 44,288 firm-year observations associated with 7,520 firms.
- **Firm universe.** U.S.-incorporated nonutility and nonfinancial firms.
- **Segment requirements.** Each industrial segment must have at least $20 million in sales.
- **Sales reconciliation.** Firm-years are excluded if total industrial or geographic segment sales do not reconcile within one percent of total firm sales.

4.2 Industrial diversification measures

A firm is industrially diversified if it reports more than one industrial business segment on Compustat's Industry Segment file.

The paper uses three industrial-diversification measures:

- a dummy for whether the firm is multisegment;
- the number of reported industrial segments; and
- a sales-based Herfindahl index,

$$H_i = \sum_s \left(\frac{Sales_{is}}{Sales_i} \right)^2 ,$$

where lower values indicate more diversification across segments.

4.3 Global diversification measures

A firm is globally diversified if Compustat's Geographic Segment file reports any foreign sales for the firm. Export sales by a domestic subsidiary are not counted as foreign operations in the main definition.

The paper uses two global-diversification measures:

- a dummy for whether the firm has foreign operations; and
- the fraction of total sales generated by foreign subsidiaries.

Interpretation. The global measure is based on operating presence abroad, not merely selling products abroad. This matters because the paper is studying the expansion of firm boundaries across national markets.

4.4 Firm characteristics

The paper compares firms across four groups: single-segment domestic firms, multisegment domestic firms, single-segment multinationals, and multisegment multinationals. Multisegment multinationals are much larger than the other groups. Single-segment multinationals have lower leverage and higher R&D and advertising intensity. These differences motivate the multivariate controls in the valuation analysis.

4.5 Excess value

The main valuation measure is excess value. It is the log ratio of actual firm value to imputed stand-alone value:

$$EV_i = \log \left(\frac{Actual\ Value_i}{\sum_s Sales_{is} \times Median \left(\frac{Total\ Capital}{Sales} \right)_{industry(s),year}} \right) .$$

Actual value is measured as market value of equity plus book value of total assets minus book value of common equity. Imputed segment values are based on single-segment domestic firms in the same industry and year, using the narrowest SIC grouping with at least five benchmark firms.

- Positive excess value means the firm is worth more than the imputed value of its stand-alone segments.
- Negative excess value means the firm trades at a discount relative to comparable focused domestic firms.
- Extreme observations are excluded when actual value is more than four times imputed value or less than one-fourth imputed value.
- Excess values can be computed for 34,200 of the 44,288 original firm-years.

5 Methodology and empirical specifications

5.1 Diversification trends

The paper first reports annual diversification measures. It does this in two ways.

- **Full sample.** This includes all firms observed in each year. In this sample, industrial diversification falls sharply while global diversification rises moderately.
- **1984 cohort.** This starts with firms observed in 1984 and follows them through 1997. This avoids confusing true changes in strategy with entry of newer, more focused firms into the sample.

Interpretation. In the full sample, the decline in industrial diversification partly reflects the entry of new firms that are more focused. In the 1984 cohort, the fraction of firms that are industrially diversified is roughly stable, but diversified firms operate in fewer segments and with higher Herfindahl indexes.

5.2 Correlation between global and industrial diversification

The paper estimates correlations between global and industrial diversification in three ways:

- time-series correlations between firm-year levels;
- correlations between firm-year changes; and
- correlations between changes from 1984 to 1997 for firms in both years.

The core finding is that correlations are positive, not negative. This means the data do not support a simple story in which firms replace industrial diversification with global diversification.

5.3 Univariate excess-value tests

The paper partitions firms by industrial and global diversification status and reports mean and median excess values. The univariate evidence shows that globally diversified firms have negative excess value on average. It also shows that domestic multisegment firms have negative excess value. However, the raw average for multisegment multinationals is close to zero, which highlights the importance of controlling for firm characteristics.

5.4 Multivariate valuation regressions

The main regression is:

$$EV_{it} = \alpha + \beta_1 IndOnly_{it} + \beta_2 GlobalOnly_{it} + \beta_3 Both_{it} + \Gamma' X_{it} + \varepsilon_{it},$$

where the omitted group is single-segment domestic firms.

The control vector X_{it} includes relative firm characteristics measured as deviations from same-industry domestic single-segment benchmarks:

- market value of total capital;
- long-term debt to total capital;
- capital expenditures to sales;
- EBIT to sales;
- R&D to sales; and
- advertising to sales.

Interpretation. The coefficients on the diversification dummies estimate the valuation discount or premium associated with each diversification category, relative to focused domestic firms with similar industry benchmarks and controls.

5.5 Robustness tests

The paper checks whether the diversification discount is robust to several alternatives.

- **Time periods.** The global diversification discount remains stable across 1984–1988, 1989–1993, and 1994–1997.
- **Annual regressions.** Averaging coefficients across yearly regressions avoids overstated precision from pooled firm-years.
- **Firm fixed effects.** Fixed effects help control for time-invariant firm characteristics that could jointly affect value and diversification.
- **Benchmark firms.** Results are similar when imputed values use the largest same-industry single-segment firm rather than the median single-segment firm.
- **Alternative valuation ratios.** Results are similar when total capital to assets replaces total capital to sales.

- **Industry definitions.** Results are similar using two-digit rather than finer SIC matching.
- **Exports.** The global discount remains when export-intensive firms are reclassified as globally diversified or removed from benchmark portfolios.
- **Extent of foreign sales.** The discount becomes larger as the percentage of foreign sales increases.

5.6 Changes in diversification status

The paper identifies firm-years in which firms first become multinational, cease to be multinational, first become multisegment, or cease to be multisegment. It then studies the change in excess value from year -1 to year 0.

- Firms that become globally diversified experience a significant decline in excess value in the full sample.
- Firms that become industrially diversified also experience a significant decline in excess value.
- Firms that cease industrial diversification experience significant increases in excess value.
- Firms that cease global diversification experience positive changes in excess value, though the evidence is weaker and less consistently significant.

6 Identification and interpretation

6.1 What the paper can identify

The paper provides strong descriptive and regression evidence that global diversification is associated with lower value relative to domestic focused firms. It also shows that within-firm changes in diversification status line up with changes in value in the expected direction.

The evidence is not a randomized experiment. It identifies robust associations, supported by fixed effects, alternative benchmarks, and event-style changes, rather than a fully structural causal parameter.

6.2 Endogeneity concern

A central concern is that lower-valued firms may choose to diversify, or diversified firms may acquire lower-valued assets. The paper discusses this concern in light of the industrial diversification literature.

The paper's own evidence partly addresses the issue:

- firm fixed effects reduce concerns about time-invariant omitted firm characteristics;
- year-by-year and alternative benchmark regressions show robustness;
- firms that become globally diversified often have positive excess values before diversification, which weakens the simple story that low-value firms choose to diversify;
- excess value declines when firms become globally or industrially diversified, and rises when firms refocus.

6.3 Benchmark construction concern

The excess-value approach compares diversified firm segments to domestic single-segment benchmark firms. This raises several issues.

- Foreign segment country and industry details are limited in Compustat's Geographic Segment file.
- International accounting standards and investor protection differences could affect valuation ratios across countries.
- Segments of multinational firms are often larger than the median domestic benchmark firm.

The paper responds with alternative benchmark and valuation specifications, and the main negative global diversification coefficient remains.

6.4 Interpretation of the diversification discount

The paper interprets the discount as consistent with the view that the costs of global diversification outweigh the benefits for the average firm.

Possible costs include:

- inefficient internal capital allocation;
- weaker monitoring of distant managers;
- coordination costs across countries;
- managerial pursuit of private benefits from size and complexity.

The analogy is to the industrial conglomerate discount: a strategy can be popular and feasible while still failing to create value on average.

7 Tables

7.1 Table I: Measures of global and industrial diversification

Read:

- The full sample contains 44,288 firm-years.
- 17.8% of firm-years are industrially diversified, while 30.6% are globally diversified.
- Among globally diversified firm-years, 29.0% are also industrially diversified.
- Globally diversified firms derive 27.4% of sales from foreign operations on average, conditional on being globally diversified.

Economic message: Global diversification is more common than industrial diversification in the sample. The two categories overlap substantially, but they are not identical.

7.2 Table II: Firm characteristics

Read:

- Multisegment multinationals are by far the largest firms, with median total capital of about $1.43 billion.
- Single-segment multinationals have lower leverage and higher R&D intensity than domestic single-segment firms.
- Firm characteristics vary systematically across diversification groups, motivating multivariate controls.

Economic message: Raw comparisons of excess value can be misleading because diversified and multinational firms differ strongly in size, leverage, investment, and intangible-intensity proxies.

Table I

Measures of Global and Industrial Diversification

Mean and median measures of industrial and global diversification for 44,288 firm-years over the period 1984–1997.

	All Firm-Years (n = 44,288)		Industrially Diversified (n = 7,905)		Globally Diversified (n = 13,530)	
	Mean	Median	Mean	Median	Mean	Median
Industrial diversification						
Fraction of firm-years industrially diversified	0.178	N.A.[a]	1.000	N.A.	0.290	N.A.
Number of segments	1.331	1.000	2.853	3.000	1.583	1.000
Sales-based Herfindahl index	0.915	1.000	0.526	0.516	0.856	1.000
Global diversification						
Fraction of firm-years globally diversified	0.306	N.A.	0.496	N.A.	1.000	N.A.
Fraction foreign sales	0.084	0.000	0.123	0.000	0.274	0.230

[a] N.A.: Not applicable.

Table I: Measures of global and industrial diversification.

Table II

Firm Characteristics

Descriptive statistics on various firm characteristics for the sample of 44,288 firm-year observations over the period 1984–1997. The sample is partitioned into four groups on the basis of whether the firm is industrially or globally diversified in the given firm-year. Firms are classified as multisegment (i.e., industrially diversified) if they report more than one business segment on Compustat's Industry Segment File. Firms are classified as multinational (i.e., globally diversified) if Compustat's Geographic Segment File reports any foreign sales for the firm. R&D and advertising expenses are set to zero if missing. Market value of total capital is defined as the market value of equity plus the book value of total assets minus the book value of equity. Means are reported with median values in italics below.

Firm Characteristic	Single-segment Domestic	Multisegment Domestic	Single-segment Multinational	Multisegment Multinational
Market value of	672.385	1801.495	1526.708	4784.908
total capital ($mill.)	*125.974*	*456.303*	*258.654*	*1427.741*
Long-term debt/	0.162	0.193	0.116	0.163
total assets	*0.111*	*0.169*	*0.063*	*0.139*
EBIT/sales	0.122	0.119	0.121	0.136
	0.097	*0.099*	*0.120*	*0.125*
Capital expenditures/	0.097	0.068	0.072	0.064
sales	*0.038*	*0.038*	*0.043*	*0.046*
R&D/sales	0.022	0.006	0.054	0.021
	0.000	*0.000*	*0.023*	*0.011*
Advertising/sales	0.012	0.010	0.015	0.015
	0.000	*0.000*	*0.000*	*0.000*

Table II: Firm characteristics by diversification group.

7.3 Table III: Annual industrial and global diversification measures

Read:

- In the full sample, industrial diversification falls from 0.263 in 1984 to 0.123 in 1997.
- In the full sample, global diversification rises from 0.292 to 0.329.
- In the 1984 cohort, global diversification rises sharply from 0.292 to 0.449.
- Among globally diversified firms in the 1984 cohort, foreign sales rise from 21.1% to 29.1% of sales.

Economic message: Global diversification became more prevalent and more intensive over time. The decline in industrial diversification is most pronounced in the full sample, partly because newer entrants are more focused.

7.4 Table IV: Correlations among global and industrial diversification

Read:

- Global diversification is positively correlated with being multisegment.
- Changes in global diversification are positively correlated with changes in industrial diversification.
- The 1984–1997 change correlations are also generally positive.

Economic message: The paper finds no evidence that firms substitute global diversification for industrial diversification. The two strategies appear closer to complements than substitutes.

Table III

Average Annual Industrial and Global Diversification Measures

The sample includes 44,288 firm-years over the period 1984–1997. The number of industrial segments and Herfindahl measures are for the subsample of firms that are industrially diversified. The fraction foreign sales measure is for the subsample of firms that are globally diversified. The 1984 sample includes those firms for which there is Compustat data available in 1984. We then follow this set of firms between 1984 and 1997.

			Industrially Diversified Firm-Years Only			Globally Diversified Firm-Years Only
Year	N	Fraction Industrially Diversified	Number of Industrial Segments	Sales-based Herfindahl	Fraction Globally Diversified	Fraction Foreign Sales
Panel A: Full Sample						
1984	2,356	0.263	3.176	0.486	0.292	0.211
1985	2,522	0.244	3.110	0.497	0.284	0.220
1986	2,643	0.233	2.992	0.506	0.288	0.242
1987	2,771	0.216	2.940	0.512	0.287	0.256
1988	2,752	0.207	2.907	0.525	0.289	0.271
1989	2,749	0.199	2.875	0.528	0.293	0.277
1990	2,807	0.192	2.846	0.533	0.300	0.291
1991	2,925	0.182	2.765	0.536	0.304	0.280
1992	3,161	0.170	2.784	0.537	0.308	0.284
1993	3,391	0.161	2.744	0.537	0.302	0.278
1994	3,658	0.150	2.717	0.538	0.312	0.274
1995	4,084	0.138	2.706	0.542	0.319	0.291
1996	4,259	0.130	2.656	0.549	0.331	0.295
1997	4,210	0.123	2.606	0.557	0.329	0.296
Panel B: 1984 Sample						
1984	2,356	0.263	3.176	0.486	0.292	0.211
1985	2,059	0.254	3.134	0.494	0.303	0.215
1986	1,872	0.256	3.031	0.504	0.324	0.240
1987	1,719	0.250	3.016	0.506	0.338	0.251
1988	1,568	0.250	2.954	0.517	0.344	0.272
1989	1,421	0.253	2.919	0.523	0.353	0.278
1990	1,337	0.258	2.948	0.529	0.361	0.293
1991	1,275	0.260	2.870	0.534	0.372	0.287
1992	1,227	0.264	2.895	0.532	0.394	0.288
1993	1,180	0.264	2.836	0.536	0.401	0.289
1994	1,131	0.261	2.820	0.540	0.409	0.288
1995	1,063	0.262	2.860	0.538	0.427	0.301
1996	1,022	0.267	2.795	0.541	0.443	0.299
1997	940	0.256	2.759	0.543	0.449	0.291

Table III: Annual industrial and global diversification measures.

Table IV

Correlations among Measures of Global and Industrial Diversification

The sample includes 44,288 firm-years between 1984 and 1997. Panel A presents the average of the time-series correlation between firm-year levels of global and industrial diversification, Panel B presents correlations among changes in diversification in individual firm-years, and Panel C presents correlations among changes in diversification between 1984 and 1997 for the subsample of firms that are included in the sample in both 1984 and 1997.

Panel A: Time-series Correlation between Firm-Year Levels of Global and Industrial Diversification		
	Multisegment Dummy	Number of Segments
Global dummy	0.309***	0.203***
% foreign sales	0.055	−0.014
Panel B: Correlations between Firm-Year Changes in Global and Industrial Diversification		
	Δ Multisegment Dummy	Δ Number of Segments
Δ Global dummy	0.059***	0.064***
Δ % foreign sales	0.017***	0.020***
Panel C: Correlations Between Changes in Global and Industrial Diversification between 1984 and 1997		
	Δ Multisegment Dummy	Δ Number of Segments
Δ Global dummy	0.070**	0.074**
Δ % foreign sales	0.030	0.028

***, ** Significant at the 0.01 and 0.05 levels, respectively.

Table IV: Correlations between global and industrial diversification.

7.5 Table V: Excess value measures

Read:

- Globally diversified firms have mean excess value of -0.034 and median excess value of -0.047.
- Multisegment firms have mean excess value of -0.026 and median excess value of -0.035.
- Single-segment multinationals show a clear discount; domestic multisegment firms also show a discount.
- Multisegment multinationals do not show a significant raw discount in the univariate table, which motivates multivariate analysis.

Economic message: The univariate evidence already points to a global diversification discount, but firm characteristics confound the raw comparison.

7.6 Table VI: Multivariate regression tests

Read:

- Relative to focused domestic firms, the discount is -0.204 for firms that are only industrially diversified.
- The discount is -0.182 for firms that are only globally diversified.
- The discount is -0.322 for firms that are both industrially and globally diversified.
- The global diversification discount is stable across subperiods.

Economic message: Once firm characteristics are controlled for, global diversification is associated with a sizable and statistically strong valuation discount similar to the industrial diversification discount.

Table V

Excess Value Measures for a Sample of 34,200 Firm-Years between 1984 and 1997

From the original sample of 44,288 firm-years, we exclude those for which there is insufficient data to calculate excess value. We also exclude outlier observations, defined as those observations for which actual value is either more than four times imputed value or less than one-fourth imputed value. Excess value is measured as the log of the ratio of the firm's total market value to the sum of the imputed market values of its segments. Imputed segment values are calculated by multiplying the median ratio of total capital to sales for single-segment domestic firms in the same industry times the level of sales for the individual segment. Means are reported above medians, with the number of observations in parentheses below. The sample is partitioned by whether or not the firm is industrially diversified, and whether or not the firm is globally diversified in a given firm-year. Firms are classified as multisegment (i.e., industrially diversified) if they report more than one business segment on Compustat's Industry Segment File and as single-segment otherwise. Firms are classified as multinational (i.e., globally diversified) if Compustat's Geographic Segment File reports any foreign sales for the firm and as domestic otherwise. Significance of means and medians are measured using a standard two-tailed t-test and a two-tailed Wilcoxon signed rank test, respectively.

	Single-segment	Multisegment	All
Domestic only	−0.000	−0.059***	−0.007**
	0.000	−0.057***	0.000***
	(n = 20,278)	(n = 2,881)	(n = 23,159)
Multinational	−0.049***	0.008	−0.034***
	−0.064***	−0.010	−0.047***
	(n = 8,193)	(n = 2,848)	(n = 11,041)
All	−0.014***	−0.026***	
	0.003***	−0.035***	
	(n = 28,471)	(n = 5,729)	

***, ** Significant at the 0.01 and 0.05 levels, respectively.

Table V: Excess value measures by diversification category.

Table VI

Multivariate Regression Tests

Ordinary least squares regressions of excess value on dummy variables denoting industrial and global diversification, and a set of control variables. Excess value is measured as the log of the ratio of the firm's total market value to the sum of the imputed market values of its segments. Imputed segment values are calculated by multiplying the median ratio of total capital to sales for single-segment domestic firms in the same industry times the level of sales for the individual segment. A firm is industrially diversified if it reports more than one industrial business segment. Likewise, a firm is globally diversified if Compustat's Geographic Segment File reports any foreign sales for the firm. All control variables are measured as deviations from the median value for domestic single-segment firms in the same industry. The sample includes 34,200 firm-year observations over the period 1984–1997. Coefficient estimates are reported with t-statistics in parentheses below. Results are reported for the full sample of firm-years, the subperiods of 1984–1988, 1989–1993, and 1994–1997.

Independent Variables	Full Sample	1984–1988	1989–1993	1994–1997
Intercept	0.002	0.014	0.012	−0.014
	(0.65)	(2.26)	(2.12)	(−2.45)
Dummy equal to one if only industrially diversified	−0.204	−0.245	−0.190	−0.166
	(−21.67)	(−16.20)	(−11.91)	(−9.10)
Dummy equal to one if only globally diversified	−0.182	−0.181	−0.169	−0.184
	(−29.04)	(−15.29)	(−15.88)	(−18.05)
Dummy equal to one if both industrially and globally diversified	−0.322	−0.393	−0.284	−0.276
	(−31.27)	(−22.82)	(−15.91)	(−15.13)
Relative market value of total capital	0.127	0.119	0.123	0.134
	(67.45)	(34.06)	(38.08)	(43.26)
Relative long-term debt to total capital	−0.409	−0.235	−0.492	−0.468
	(−23.99)	(−7.58)	(−18.09)	(−15.54)
Relative capital expenditures to sales	0.415	0.356	0.315	0.486
	(26.10)	(12.55)	(9.46)	(20.53)
Relative EBIT to sales	1.041	0.885	1.394	0.997
	(46.09)	(23.42)	(30.98)	(27.54)
Relative R&D to sales	1.278	1.333	0.920	1.387
	(32.17)	(12.05)	(11.37)	(26.38)
Relative advertising to sales	0.004	0.326	−0.045	−0.192
	(0.05)	(2.25)	(−0.32)	(−1.38)
Adjusted R^2	0.267	0.231	0.291	0.283
Number of observations	33,478	9,793	10,942	12,473

Table VI: Multivariate regression tests.

7.7 Table VII: Sensitivity and robustness tests

Read:

- Annual regressions, firm fixed effects, alternative benchmark firms, alternative value measures, and two-digit industry matching all produce negative global diversification coefficients.

- The global-only coefficient ranges from -0.120 to -0.203 across these robustness specifications.
- The both-global-and-industrial coefficient remains negative in all specifications.

Economic message: The global diversification discount is not an artifact of one particular pooled specification, benchmark construction, valuation ratio, or industry definition.

7.8 Table VIII: Excess values and changes in diversification

Read:

- Firms that become globally diversified have positive excess value in year -1, then experience a mean change in excess value of -0.077 in year 0.
- Firms that become industrially diversified experience an even larger mean change of -0.203.
- Firms that cease industrial diversification experience a large positive mean change of 0.220.
- Ceasing global diversification is associated with positive changes, but the full-sample estimates are less statistically precise.

Economic message: The change analysis reinforces the cross-sectional evidence. Entry into diversification is associated with value declines, while refocusing is associated with value increases.

Table VII

Sensitivity and Robustness Tests

Alternative regressions of excess value on dummy variables denoting industrial and global diversification, and a set of control variables. Model (1) reports mean coefficient estimates from annual ordinary least squares regressions. Model (2) reports estimates from a fixed-effects model. Model (3) computes imputed segment values by multiplying the ratio of total capital to sales for the largest (in sales) single-segment domestic firm in the same industry times the level of sales for the individual segment. Model (4) computes imputed values using the ratio of total capital to assets rather than total capital to sales. Model (5) defines industry at the two-digit, rather than four-digit level. A firm is industrially diversified if it reports more than one industrial business segment. Likewise, a firm is globally diversified if Compustat's Geographic Segment File reports any foreign sales for the firm. All control variables are measured as deviations from sales-weighted industry median values. The sample includes 34,200 firm-year observations over the period 1984–1997. Coefficient estimates are reported with t-statistics in parentheses below.

Independent Variables	Means of Annual Estimates (1)	Fixed Effects (2)	Largest Single-segment Firm (3)	Value/ Assets (4)	2-digit Industry Match (5)
Intercept	0.005	−0.045	0.113	0.067	−0.013
	(1.18)	(−9.89)	(22.88)	(24.45)	(−3.74)
Dummy equal to one if only industrially diversified	−0.199	−0.245	−0.210	−0.150	−0.197
	(−17.46)	(−17.47)	(−16.48)	(−5.45)	(−20.48)
Dummy equal to one if only globally diversified	−0.177	−0.203	−0.120	−0.145	−0.148
	(−25.12)	(−19.13)	(−15.55)	(−28.07)	(−23.23)
Dummy equal to one if both industrially and globally diversified	−0.316	−0.279	−0.335	−0.235	−0.308
	(−19.27)	(−16.46)	(−24.33)	(−7.30)	(−29.66)
Relative market value of total capital	0.123	0.222	0.091	0.095	0.128
	(46.26)	(65.10)	(49.01)	(56.52)	(67.89)
Relative long-term debt to total capital	−0.401	−0.411	−0.278	−0.984	−0.393
	(−8.79)	(−21.75)	(−15.87)	(−64.54)	(−23.06)
Relative capital expenditures to sales	0.422	0.340	0.416	−0.004	0.459
	(9.83)	(20.41)	(23.87)	(−0.35)	(28.62)
Relative EBIT to sales	1.150	0.455	0.808	0.462	1.158
	(13.86)	(18.04)	(35.68)	(27.01)	(51.03)
Relative R&D to sales	1.281	0.518	0.961	0.426	1.582
	(12.57)	(9.50)	(25.27)	(14.58)	(39.54)
Relative advertising to sales	0.058	−0.155	−0.370	0.454	0.448
	(0.73)	(−1.18)	(−4.09)	(6.29)	(5.61)
Adjusted R^2	0.272	0.656	0.210	0.256	0.295
Number of observations	14	33,478	26,209	28,895	33,163

Table VII: Sensitivity and robustness tests.

Table VIII

Excess Values and Changes in Diversification

Year −1 levels and the change in excess value from year −1 to year 0 are presented. Year 0 is the year in which the firm changes its diversification status. Excess value is measured as the log of the ratio of the firm's total market value to the sum of the imputed market values of its segments. Imputed segment values are calculated by multiplying the median ratio of total capital to sales for single-segment domestic firms in the same industry times the level of sales for the individual segment. Firms are classified as multisegment (i.e., industrially diversified) if they report more than one business segment on Compustat's Industry Segment File and single-segment otherwise. Firms are classified as multinational (i.e., globally diversified) if Compustat's Geographic Segment File reports any foreign sales for the firm and as domestic otherwise. Panel A presents results for the full sample, while panels B, C, and D present results for the 1984–1988, 1989–1993, and 1994–1997 subperiods. Significance of means and medians are measured using a standard two-tailed test and a two-tailed Wilcoxon signed rank test, respectively.

		Year −1 Excess Value		Δ Excess Value	
	N^a	Mean	Median	Mean	Median
Panel A: Full Sample					
Became globally diversified	511	0.126***	0.102***	−0.077***	−0.056***
Ceased global diversification	150	−0.047	0.009	0.067	0.055
Became industrially diversified	147	0.042	0.005	−0.203***	−0.198***
Ceased industrial diversification	256	−0.085***	−0.087***	0.220***	0.156***
Panel B: 1984–1988					
Became globally diversified	128	0.062	0.019	−0.010	0.046
Ceased global diversification	42	−0.091	−0.037	0.193**	0.102*
Became industrially diversified	39	0.013	0.016	−0.174**	−0.127**
Ceased industrial diversification	86	−0.148***	−0.222***	0.201***	0.103***
Panel C: 1989–1993					
Became globally diversified	166	0.161***	0.167***	−0.084***	−0.088***
Ceased global diversification	49	−0.035	0.049	0.009	0.007
Became industrially diversified	48	−0.035	−0.001	−0.243***	−0.214***
Ceased industrial diversification	74	−0.065	−0.076	0.229***	0.206***
Panel D: 1994–1997					
Became globally diversified	217	0.136***	0.104***	−0.111***	−0.091***
Ceased global diversification	59	−0.025	−0.026	0.026	0.067
Became industrially diversified	60	0.131**	0.029*	−0.191***	−0.189***
Ceased industrial diversification	74	−0.065	−0.076	0.229***	0.206***

[a] This represents the number of firms for which we are able to calculate a change in excess value.

***, **, * Significant at the 0.01, 0.05, and 0.10 level, respectively.

Table VIII: Excess values and changes in diversification.

8 Short summary

- The paper studies global and industrial diversification among U.S. firms from 1984 to 1997.
- Global diversification is common: 30.6% of firm-years report foreign operations.
- Global diversification increases over time, especially among firms already present in 1984.
- Industrial diversification generally declines in the full sample, but firms do not appear to substitute global diversification for industrial diversification.
- The two forms of diversification are positively correlated in levels and changes.
- Globally diversified firms trade at a discount relative to comparable single-segment domestic firms.
- In the main multivariate regressions, the global-only discount is about 0.18, close to the industrial-only discount of about 0.20.
- Firms that are both globally and industrially diversified have the largest discount, about 0.32.

- The results are robust to time-period splits, annual regressions, firm fixed effects, alternative benchmarks, alternative value measures, and alternative industry definitions.
- Entry into global or industrial diversification is associated with declines in excess value; refocusing is associated with increases in excess value.

9 Conclusion

9.1 Core conclusion

Denis, Denis, and Yost show that global diversification increased among U.S. corporations over 1984–1997, but that this increase did not create value for the average firm. Globally diversified firms trade at significant valuation discounts relative to comparable focused domestic firms.

9.2 Economic intuition

Global diversification can create value through intangible assets, flexibility, and access to international markets. But it can also destroy value through coordination costs, monitoring problems, inefficient internal capital allocation, and managerial private benefits. The paper's evidence suggests that the average costs dominate the average benefits.

9.3 Incremental contribution revisited

The paper extends the diversification-discount literature from industrial conglomerates to multinational firms. It also shows that global diversification is not simply a replacement for industrial diversification. Rather, firms that diversify along one dimension are more likely to diversify along the other as well.

9.4 Implications

The findings caution against treating international expansion as automatically value enhancing. A firm may gain from foreign operations when it has specific assets or opportunities, but global diversification as a broad corporate strategy does not appear to increase value on average.

9.5 Limitations

The analysis is observational and relies on segment reporting. Geographic segment data are less detailed than industrial segment data, especially because Compustat does not fully identify countries and industries of foreign segments. The excess-value method also depends on benchmark firms and valuation ratios. The paper addresses these concerns with extensive robustness checks, but cannot completely eliminate all endogeneity concerns.

9.6 Takeaway

The enduring takeaway is that global diversification resembles industrial diversification more than it differs from it. Both became plausible corporate strategies, both can offer managers private benefits, and both are associated with valuation discounts for the average firm.